

Walking: The benefit of being on two legs

James Earls

Introduction

Many palaeontologists have a rare point of agreement on the importance of upright gait as a major, if not the main, impetus for evolution of the *Homo sapiens* lineage (Lieberman, 2012). Obligate bipedalism was one of the most significant evolutionary developments as locomotor economy allowed the diversion of metabolic resources to “expensive tissues”, such as the brain. However, despite new fossil finds, increasing clarity of comparative anatomy and better research all round, the origins of bipedalism are still not yet fully understood (Osborn, 1928; Lieberman, 2012; Pontzer, 2017). The fossil record indicates bipedal adaptations probably began 6–7 million years ago (MYA), but it was not until around 2 MYA that locomotor efficiency was fully developed through a suite of anatomical refinements (Pontzer, 2017).

This chapter will blend comparative and functional anatomy with paleontological research to provide a unifying understanding of why and how our ancestors made the change from some form of quadrupedalism to our unique upright bipedal locomotion. Such an approach should provide a better appreciation of whole-body mechanics and the intricate interaction between form and function for various tissue types. Finally, it aims to illustrate some new ideas on clinical reasoning for movement dysfunctions and various pathologies.

Efficiency

There is general consensus that upright walking is metabolically cheaper than staying down on all four limbs (Figure 30.1). In a world where calories were the only currency, being able to move between food sources, shelter, mating potentials and safety with as little metabolic cost as possible confers significant evolutionary advantage. Sockol et al. (2007) (Figure 30.1) compared the cost of locomotion of a number of individual chimps and humans. In each case, the humans were significantly more efficient in their gait regardless of whether the chimps walk on two or four limbs. Human bipedalism was found to be up to 75% less costly in comparison.

While human fascial tissues are undoubtedly involved and part of the reason for economical propulsion, we must first explore some of the other contributing factors that facilitate their exploitation.

Skeletal changes

Aligning the center of gravity directly above our feet is not as simple as just standing up. Our vertical stability is facilitated by the lumbar lordosis and posteriorly extended ischial tuberosities that provide leverage for the hamstrings even when the hip is extended. As seen in Figure 30.1, chimps do not fully straighten their lower limbs when adopting a bipedal posture and have to

Chapter 30

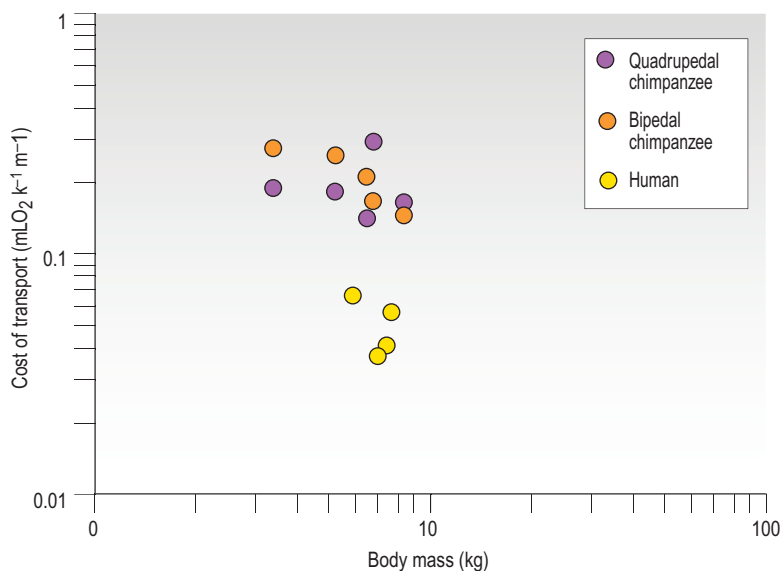


Figure 30.1

Net cost of transport (ml of O₂ kg⁻¹ m⁻¹) for chimpanzee quadrupedal walking (purple), chimpanzee bipedal walking (orange) and human walking (yellow).

Adapted from Sockol et al. (2007).

maintain bent knees and hips. The chimps' lower limb flexions are not due to limited range in their hips or knees, they are necessary because of limited lumbar extension and reduced support from the hamstrings when upright due to an inferiorly extended ischial tuberosity.

Limited space prevents a fuller exploration of each plane and nuance of gait but we should at least acknowledge the abductors' contribution to frontal plane stability. In contrast to our primate cousins, our ilia face laterally allowing the gluteals to provide frontal plane stability during single leg support. Being able to support our center of mass during single-leg stance without pitching our trunk from side-to-side allows us to move forward more smoothly. Frontal plane support provided by the gluteals resulted from the most notable shape change in the pelvis as it evolved to turn the gluteals from backward-facing extensors to lateral stabilizers by changing the orientation of the iliac blades.

Lateral stability and our upright stance facilitate a longer stride but only when coupled with

parasagittal range from the ankle and toes. Taking fewer strides to cover a distance is beneficial and many adaptations occurred in the foot. The foot morphed from a climbing tool and became an adaptable rigid lever to aid efficient walking. To assist a long stride the great toe changed from its adducted grasping position and toe extension increased at each metatarsal phalangeal joint (MTP). Both modifications, alignment of the hallux and increased MTP extension, were coupled with the development of an arched foot. The intricate multi-joint arrangement of the foot created a locomotor base that can be stable at heel-strike, unlock and adapt to the surface while absorbing ground reaction forces during stance phase.

During the gait cycle, the human foot experiences two peaks of ground reaction forces, one following heel-strike and the other towards toe-off. It appears advantageous to have a skeletally stable foot at these points and a mobile foot in between which allows some force to be absorbed by the plantar tissues and facilitates the foot's adapta-

Walking: The benefit of being on two legs

tion to the substrate. The half-domed arch of the human foot allows the foot to alter its state from being more stable (supinated) to relatively mobile (pronated) and return to a stable, supinated condition in readiness for the release of energy toward toe-off (McKeon et al., 2014) (Figure 30.2A).

Re-supination of the foot prior to toe-off is enhanced by MTP extension (Fernández et al., 2018). As the toes extend, the plantar tissues (plantar fascia, long and short toe flexors) are tensioned to draw the foot's many bones together. Tensioning of the plantar tissues forms part of the “windlass mechanism” that stiffens the foot in preparation for propulsion at toe-off (Holowka and Lieberman, 2018). The supinated foot provides the stable platform for the release of energy and has formed as a result of the parasagittal progression over the foot; deviating to either side will negatively affect supination.

Parasagittal movement over the foot is facilitated by the adducted great toe and toe extension changes discussed above. We fail to see these two peaks of reaction forces in the gait pattern of other primates (Figure 30.3). Our ape cousins lack the functional combination of an adaptable rigid lever, adducted great toe and enhanced toe extension lack and hence detour into other planes during stance phase.

The net result of the skeletal changes was that *Homo* could expend less energy working against gravity and, as we explore below, could capture and recycle kinetic energy from momentum. Coupled hip and low back extension and an adaptable foot allowed the development of a longer, straight-legged stride. Using four rockers with the foot and ankle complex—heel, ankle, forefoot and toe—allows a straight-leg heel-strike and for the trunk to then vault over a dynamically reactive foot. The new skeletal alignment and joint ranges reduced the need for muscle work to

resist gravity and deal with impact forces but also brought new possibilities in how humans (*Homo sapiens*) could use their soft tissues.

Soft tissue mechanics

The locomotor system incorporates elastic soft tissue elements that exchange energy within the environment. Recruitment of elastic elements provides an efficiency which is enhanced by the increased joint ranges outlined above. The greater joint range allows more strain of the elastic tissues, which again profits locomotor efficiency through a number of mechanisms outlined below.

Muscle and the deep fascia that holds it in place form a symbiotic relationship that can optimize efficiency during movement. Various researchers have shown the interplay between in-series and in-parallel collagenous tissues and the contractile tissues when experiencing repeated rhythmical movement and how the system optimizes to reduce eccentric and concentric and concentric muscle contractions (Fukunaga et al., 2002; Sawicki et al., 2009). For example, the in-series tendons of the plantar flexor group uses the body's momentum as it “vaults” over the foot to correct and bring the foot out of pronation (the windlass effect) while the tendons simultaneously lengthen and the muscles isometrically contract. While it is necessary to understand the properties and resultant interactions between tissues of varying stiffness (Fukunaga et al., 2002), doing so is extremely difficult because of the problems in separating the energy contribution of one from the other (Roberts, 2016).

As seen above, toe extension facilitates the windlass mechanism of the foot by tensioning of the plantar tissues and toe flexors. The plantar flexors and toe flexors are tensioned neurologically and by stretch during the latter moments

Chapter 30

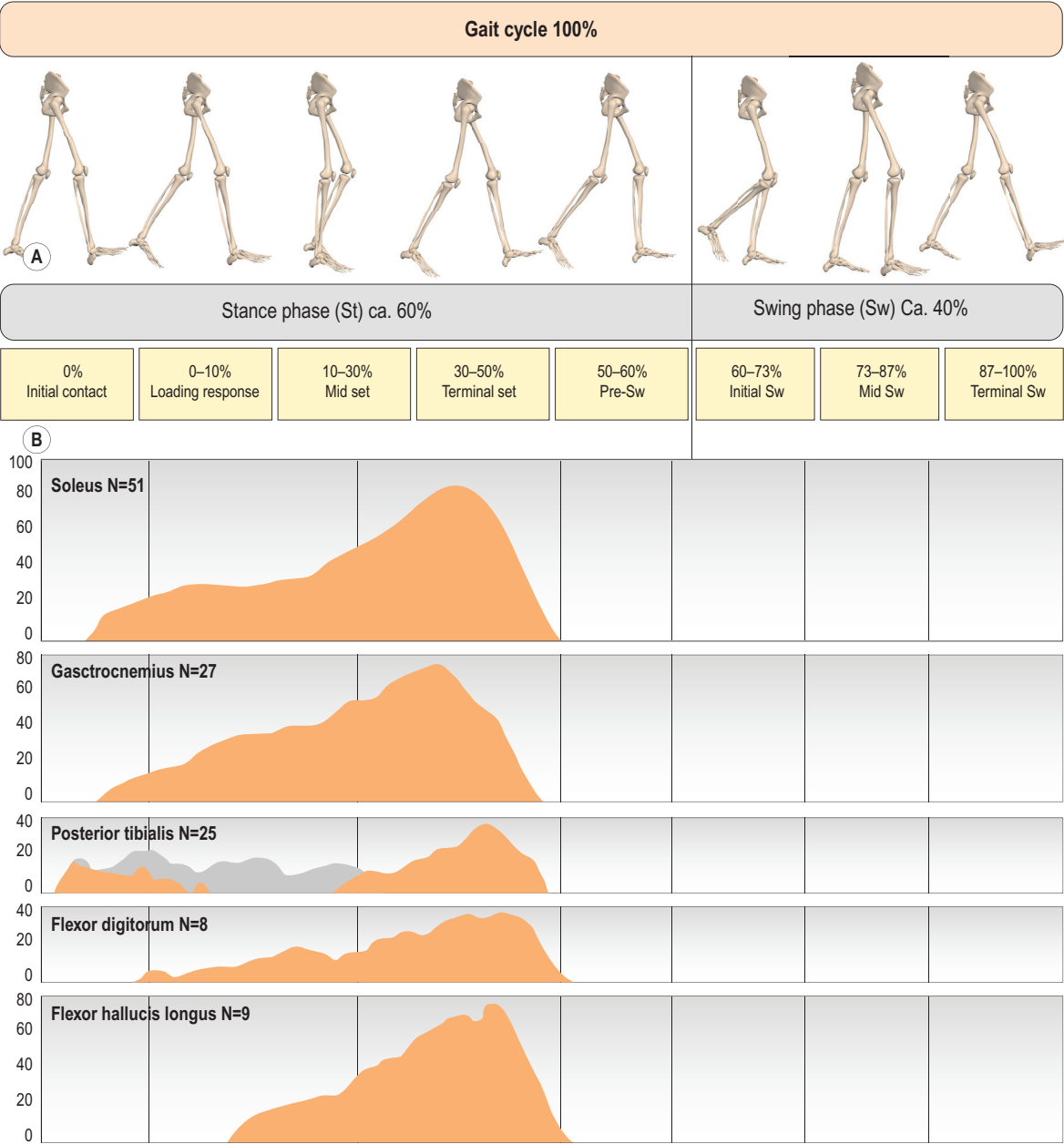


Figure 30.2

(A) Gait cycle phases and percentage portion of each. Heel strike is represented as “initial contact” and occurs at 0% and 100%, toe-off occurs at 60%. **(B)** Quantified electromyography readings are shown as a percentage of maximum manual muscle test value (dark shading) and are aligned to the relevant phase of gait.

Walking: The benefit of being on two legs

of stance phase as the ankle dorsiflexes and the toes extend (see Figures 30.2A and B, mid- to late-stance). The body's forward momentum has to be decelerated by the plantar soft tissues, which tend to be pennate muscles. By arranging shorter muscles fibers at angles, pennated (feathered) muscle architecture provides maximum force production for the least amount of volume.

Pennation is an important feature of distal limbs muscles as it reduces their volume without sacrificing potential force output. Feet and legs move furthest during locomotion and reducing inertia pennation provides another energy-saving design (McNeill Alexander, 2002; Biewener, 2016; Roberts, 2016). Interestingly, pennation may have been selected for in our ancestry purely as a weight-saving device (Biewener, 2016), but other primates do not appear to have unlocked the full elastic potential of this arrangement.

Utilization of elastic energy benefits from a number of anatomical features and is enhanced by the correct myofascial architecture: pennated muscles with long in-series tendons. There must also be enough momentum involved to strain the tendons an appropriate degree, but not so much momentum that the muscles cannot control it. Achieving tendon stretch also requires associated joints to have a range of movement that allows lengthening of the tissues, hence why the development of a long stride and its associated joint extensions are so important.

Muscles act eccentrically to control movement, commonly in the opposite direction of their listed "actions". We saw this above as part of the windlass mechanism: the plantar and toe flexors are tensioned by dorsiflexion and toe extension. This apparent reversal of function can initially make EMG readings confusing but in Figures 30.2A and B we can see that the plantar

flexors are stimulated to control the lower leg as it progresses over the foot into dorsiflexion from mid- to late stance prior to toe-off.

Muscle force combined with momentum will lengthen, or strain, tendons, and it is the strain that provides the mechanism to capture, store and then release kinetic energy. Tendons can strain to 5%; failure is said to occur at 7–12% of strain (although it often experiences more), and it can recover 97% of the energy used to stretch it (McNeill Alexander, 2002). One can sometimes experience this build-up of energy when it is inadvertently released walking over a patch of ice or sand on a pavement. Loss of friction for the foot as it extends gives a premature release of captured energy and the foot flicks rapidly backwards before reaching full toe-off position.

It is this author's belief that captured energy is held within the system during gait because of friction between foot and ground, and as the body progresses over the foot into the series of extensions a catapult-like mechanism is created. Catapult and trigger mechanisms are well-known within the animal kingdom, especially for ballistic feeders and high jumpers like the flea, but each of these usually have an obvious locking mechanism that allows the build-up of muscle tension to lengthen the elastic springs (Vogel and DeFerrari, 2013). Ballistic feeders and athletic jumpers of the animal kingdom release the locks to unleash the energy and, I believe, we roll over our toes to achieve the same end.

As the collagenous elastic tissues capture energy the muscles associated with any tendon can be contracting eccentrically, concentrically or isometrically, depending on the associated forces and the stiffness inherent within the tendon. For example, if the tendon is more compliant and momentum is high the muscle may

Chapter 30

need to shorten to control the movement and, conversely, if the tendon is stiff the muscle will need to lengthen to allow an appropriate range of movement to occur. Efficient loading of calf musculature requires enough range of the ankle and toe joints but being able to take a long enough stride to facilitate the stretch also recruits the ranges of the same-side knee, hip and lumbar spine (Whittington et al., 2008).

Loss of extension in any one of these joints (toes, ankle, knee, hip or lumbar spine) will influence the extension and/or the alignment of the others. A simple experiment of walking with a restriction in one of them will reaffirm the important interrelationships through the body: we must be able to couple each of those joint extensions

simultaneously to maximize the efficiencies within the myofascial system. In performing the experiment one often feels the extra workload placed on the musculature around the hip and low back. The loss of range into extension creates a need for more active concentric positive work.

The hip flexors tension during hip extension as the limb progresses into late stance toward toe-off, thereby providing deceleration and control of the movement while also loading elastic strain into the anterior hip region (Figure 30.3). Although the hip does not have the same collection of long tendons available to the foot and ankle, it has been calculated that up to 40% of the net negative work during extension is performed by passive elastic tissues (single and two-joint

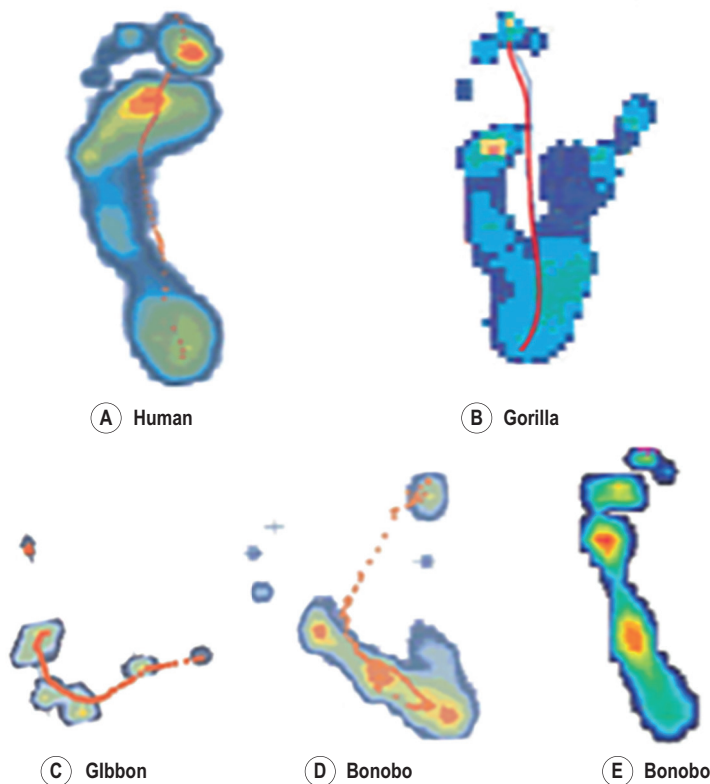


Figure 30.3 Pressure pad readings from a number of primate species.

(A) Human shows a heel strike followed by lateral to medial translation of body weight with an overall parasagittal progression. Two peaks of pressure are seen, one at heel strike and another at toe-off.

(B) Gorilla pressure readings show a wide distribution with only small peaks in the mid/fore-foot region.

(C) The gibbon shows a change of almost 90 degrees during weight transfer.

(D & E) The bonobo demonstrates two general strategies, one with a strong lateral to medial transfer, and the other (E) a roll over the lateral aspect of the foot. Peak pressures are seen through the mid-foot in both cases.

Adapted from Crompton, Vereecke and Thorpe, 2008

Walking: The benefit of being on two legs

muscles, joint capsule, ligament and skin; Silder et al., 2007), and that they provide ~50% of the net positive work during the hip flexor power burst during toe-off (Whittington et al., 2008).

Achieving the idealized toe-off position (seen at 60% of gait cycle in Figure 30.3) requires the interplay between a number of joint ranges and tissue tensioning. Immediately prior to toe-off, the toes, knee, hip and lumbar spine are all extended. The increased range at each joint has loaded an amount of energy into the fascial tissues and they are all ready to release that load at toe-off. The amount of passive energy being produced is difficult to calculate and figures vary within the literature. There are many estimates for the contribution of the Achilles tendon, but individual contribution is almost meaningless in a system of this complexity as many fascial tissues contribute to efficiency of human gait.

Muscle force and timing

Just as joint ranges are interdependent during gait, so too is the interaction between muscle and fascial tissue length. Muscle-tendon mechanics were investigated by Rubenson et al. (2012) and fascicle length was found to be optimized for force production during walking and running to within a narrow range on the ascending limb of its force-length curve. Rubenson et al. concluded that the regulation of muscle strain is probably regulated by tendon stretch.

Similar conclusions are drawn from the work of Sawicki et al. (2009), who showed that elastic tissue strain appears attuned to fascicle length in a complex system that optimizes force production. This relationship is significant because force output is reduced when muscle fibers contract from shortened and lengthened positions, and there is a mid-range in which myosin and actin fibrils are optimally aligned to produce force. Maintaining the Goldilocks length of both

elastic and contractile tissues will minimize metabolic cost while optimizing force output.

Muscle force output is also affected by speed of contraction with slower speeds generally producing higher forces. It is known that elastic loading allows a slower contraction of the muscle when the load is released (Vogel, 2013). The initial momentum of the movement is produced by elastic energy released from the fascial tissues and the muscle can then contract at a speed which maximizes its force-velocity relationship, that is, a muscle contracting at speed is less able to produce force for the movement. Elastic recoil not only contributes directly to movement but also increases the efficiency within the associated muscles.

The long stride of *Homo sapiens* therefore further adds another two muscle dynamics: force-length and force-velocity ratios are both enhanced by elastic tissue stretch. If we now contrast the long stride to our primate cousins' bent knee and bent hip gait patterns, the gulf in metabolic costs of locomotion becomes clearer. Our efficiency results from a mosaic of factors that include better alignment with gravity and a long stride, an adaptable foot that allows heel-strike and toe-off, hips and lumbar spines that facilitate extension; and each range of motion assists the simultaneous loading and release of kinetic energy in a range of elastic tissues which, in turn, also optimizes muscle force output.

Cyclical movement, such as walking, provides the human movement system with a range of energy efficiencies. Many muscles, especially those which are pennated, benefit from a stretch-shorten contraction cycle (Biewener, 2016). The lengthening of the tissue activates the muscle tendon unit, which increases force development and provides for elastic capture and storage to increase the potential for work output with reduced metabolic energy consumption.

Chapter 30

Therefore, we have to assume that removing the stretch-shortening cycle reduces those benefits and their loss must be compensated for through concentric muscle work. Future research may give further insight to myofascial pain and dysfunction generated through compensatory overuse.

Efficiency versus endurance

Due to the attention given to long-distance running in recent years, it is worth noting the difference between efficiency and endurance and their role in gait. In terms of a locomotor strategy, walking is more efficient for moving our mass over long distances and incurs a much lower metabolic cost than running. Endurance running is metabolically costly but may have been beneficial in creating new, more successful hunting strategies, which helped create calorie surplus.

The ability to endurance run, while unmistakably instilling an evolutionary advantage, required physiological changes such as hair loss and sweating for heat management. Although endurance running may use a number of energy-efficient mechanisms, human running, by weight comparison, is actually less efficient than for other species.

Much of the extra cost associated with running compared with walking is due to the bent knee and hip necessary to reduce shock load. Ground impact with joint flexion reduces the mechanical advantage at these joints and increases muscle load. While elastic efficiencies can still be achieved with the correct running mechanics and tempo, the change from the elastically assisted straight leg vault of walking to elastically assisted flexed hip and knee running is a leap in metabolic cost.

The inefficiency of running does not undermine any of the endurance running claims but

emphasizes the need to understand the intricate interplay between joint and ground reaction force alignment in context with momentum and tissue dynamics. The important evolutionary element is that the net increase in calorie supply created by endurance running allowed the expensive brain tissue to develop. Importantly, we see many anatomical changes that enhance locomotor efficiency and endurance around 2 MYA, coinciding with expansion of cranial capacity and the transition to a more modern human body plan (Pontzer, 2017).

Elastic walking – a user’s guide

The late Leon Chaitow summarized the causes of soft tissue dysfunction as being due to “overuse, misuse, disuse and abuse” (Chaitow and DeLany, 2008), and it is this author’s belief that we have created numerous effective modalities for the assessment and treatment of the dysfunctions but we have yet to fully understand the complexities involved in the correct use of soft tissues. For example, as alluded to above, we need to explore the implications of restricted or deviated toe extension on hip flexors and many other similar interdependent relationships that play out in the moving body, whereby the loss of one range affects the potential efficiency of another area. In the first edition of this text, Adjo Zorn outlined his thoughts on the benefits of “elastic walking”. By following some of that advice, it is hoped we can find the correct use of our soft tissues to minimize the chances of “overuse, misuse, disuse or abuse”.

Through a series of tips, Zorn outlined several ways in which we can help maximize elastic contribution to gait; some of these follow below:

1. Walk with straight legs: heel-striking with a straight leg is our natural strategy but requires the ability to heel-strike on the

Walking: The benefit of being on two legs

forward leg and extend through the posterior limb. Our calcanei have adapted to absorb the shock at impact with an appropriate arrangement of the trabeculi and the natural curve of the heel facilitates the forward roll to enhance forward momentum. We then vault over the other rockers of the foot using a combination of elastic and muscle force.

2. Take long steps: the ability to heel-strike on a straight leg allows long steps, and long steps are required to transfer kinetic energy into the elastic tissues. The stretch tensions the tendon and assists the force output of the muscles. However, there are many factors that could shorten the stride. If one cannot heel-strike on the forward leg, ankle dorsiflex or extend the toes, knee, hip or lumbar on the side of the trailing limb, the stride will shorten and lose some stretch-shorten cycle benefits. Further, if we lose parasagittal range, we may compensate by using another less ideal movement strategy. This often recruits and exaggerates movement in another plane, which may influence the vector of stretch into the anterior hip, pelvis and lumbar.

However, stride length must be matched with the extension abilities of each joint. Simply striding out as far as one can will likely lead to compensations. For example, lack of toe extension prevents loading of the front of the hip but if one tries to overstride, the foot is likely to rotate and the effect will be to take the momentum to the inside of the hip rather than the front. In this case, if toe extension is irretrievable, some form of assisted toe rocker might be prescribed, either in the form of enhanced toe spring or custom orthoses, and these will allow load of the hip flexors.

I will fall short of biomechanical determinism in stating the potential for pain and dysfunction if stride length is challenged, however, clinically, it may be useful to consider the functional range throughout both limbs when confronted with seemingly localized issues. It is more scientifically defensible to predict that tissue efficiency will be reduced. The degree to which efficiency is affected is probably variable and may correlate to age, tissue health, and degree or style of movement compensation.

3. Pressure with the ball of the foot: Adjo Zorn and I differ in this advice. Zorn recommends pushing into the ground in preparation for toe-off. I totally agree if you need to speed up or walk uphill.

I have deliberately referred to toe-off in this chapter in preference to the more common “push-off”. As outlined above, should we get the mechanics correct during a normal-paced flat walk, we can swing our legs from a predominately elastic strategy. The active cue to push will incur metabolic cost through the concentric contraction it creates and, I believe, is a misunderstanding caused by the pressure readings, as seen in Figure 30.3. The increased pressure reading under the ball of the foot is naturally interpreted as part of push-off but it can also be the natural result of plantar flexor tensioning as the foot and ankle complex extends.

To get a sense of this, try putting the palm of both hands together in front of your trunk as if praying. Now, bring one elbow higher to create wrist extension on that side and feel the response on the opposing fingers. The simple act of lengthening the wrist and finger flexors created a flexion force on the side of the rising elbow. If your passive hand represents the ground and the extending

Chapter 30

wrist your foot, I hope you get a sense of the push that is created by the plantar flexor tendons around the pulley of the talus.

The cue to actively push could be a useful intervention for those with hypermobile first rays or lacking in supination, but could also interfere with the timing of the anterior hip. An active push-off could inhibit full extension of the metatarsophalangeal joint and thereby prevent full hip extension. The active push-off may therefore undermine but compensate for the lack of loading of the anterior hip.

4. Carry your pelvis: Zorn advocates the support of the pelvis from above using the tensioning of the abdominal tissues. It is an unfortunate side effect of anatomy that we have to speak of individual joints and tissues and rarely talk of overall, complete body patterns. Above, I have mentioned the coupling of hip and lumbar extension, and these will coincide with an overall anterior tilting of the pelvis. Multi-joint reaction is all one normal response through a continuous and necessarily reciprocal system. Our tissues are aligned to control movement and we need enough to facilitate the efficiency mechanisms mentioned above but not so much movement that it creates impingements elsewhere due to tissue weakness and lack of control. Functional movement is the balance between enough and not too much range. There are two dynamics at play—the need for the pelvis to tilt enough to stimulate the rectus abdominis (RA) and the ability of the RA to respond—in the ongoing dance between mobility and stability, and the need for enough and not too much of either.

As we progress through mid-stance (see Figure 30.2A), the hip begins to extend, the

pelvis is drawn into an anterior tilt and the RA will tension. RA and many hip flexors are often considered antagonists in their control of the pelvis: RA will posteriorly tilt it and the flexors will anteriorly tilt it. But, I believe, in gait they should be considered synergists, as they all control the overall extension through the body to create an elastic line along the front of the body.

A balance must be struck between the mobility of the ankle, the front of the hip and lumbar spine and the strength of each associated tissue needed to control those events. Ideally, they should be in harmony with one another, and it requires full clinical investigation to reveal, for example, if an overly lordotic spine is due to weak abdominals or restricted hip extension. As Zorn recalls, “true strength is not hardening; it is resilience, adaptability, stability. It is characterised by elasticity” (Rolf, 1989).

There are many such functional couplings, triplings and more through the body and it is my hope that true functional anatomy will filter further into clinical literature to enhance assessment and reasoning skills.

5. A sideline: elastic breathing: Zorn suggests uses a diaphragmatic breath and not letting the abdomen protrude forward. I can concur and encourage thinking of the whole system as one tensioning unit; if we lose tone and tension in one area it will influence elasticity in another. Full, normal breaths allow the thorax to support itself and, in turn, support the resiliency of other structures around it.
6. Move your upper body: there is considerable debate over the contribution of arm swing during gait. Perry and Burnfield (2010) describes the trunk as a “passenger”

Walking: The benefit of being on two legs

responsible only for its own postural integrity. While that integrity will assist the elasticity of the lower body, it does not assist with the image and, I hope, reality of the contribution to gait from the trunk. Trunk rotation, side flexion and extension will all contribute elasticity, it is a natural feature by which lengthening of the soft tissue can add efficiency into the system. If we lock that movement down, we have to find other sources to recoup lost potential energy.

Along with the tensioning (but not the prevention of movement) of the thorax through the breath, the side flexion, extension and rotations that travel up through the trunk from the pelvis during gait will tension the intercostals, abdominals and spinal tissues. Each joint movement and separation of bones created by the body's natural response to the forces traveling through it has the potential, not to create more work for the body but, judged on the work focused on lower limb tissues, should actually increase efficiency of the moving body when it is considered as a whole and not separated into units, as suggested by Perry and Burnfield.

7. The arms are important, too. Contralateral arm swing is part of the natural reaction during gait and, by creating a counterbalancing rotation, helps create tension through the system. The counter-swing of the shoulder complex seems to contain the rotations from the pelvis to cause a wringing effect that appears oriented to the oblique tissues of the abdominals and intercostals.

Of particular interest to Zorn is the more superficial relationship from one latissimus dorsi through the thoracolumbar fascia to the opposite gluteus maximus. By tensioning this

oblique line he suggests we create a sling to capture energy across the back of the trunk and pelvis and help propel it forward. This “posterior oblique sling” described by Vleeming et al. (2012) is one of many long chains of tissues that may be tensioned during and thereby assist with gait.

If we take a moment to analyze the woman in Figure 30.4 (Zorn, 2015), we could spend considerable time mapping out lines of tension and stretch that have been created by the extensions on one side and the flexion on the other. This woman is making the best of her “resonant elastic system” (Snyder and Foley, 2011 in Roberts, 2016) as she progresses forward.

Summary

The full body tensioning we see, has been allowed by the anatomical changes made over time, which have allowed movement at one joint to couple into another. The long strides with straight legs are the outcome of a number of skeletal changes in some form of primate ~7 MYA. Alterations of joint angles allowed numerous joints to be simultaneously extended to stretch the tissues crossing each of them prior to toe-off, whereby they contribute to the forward swing of the leg. The changes in joint orientation and range have led to the evolution of a metabolically efficient biped. *Homo sapiens*' optimization of fascial tissues to capture kinetic energy diverted calorie usage from locomotion to feed a larger brain.

We have focused on the back leg as it prepares for toe-off but similar, less well researched dynamics will be at work for the forward leg as it swings through to heel-strike. The lively propulsion captured in Figure 30.4 results from joints ranges developed through our hominin lineage, and those interrelated ranges grant us an ability

Chapter 30



Figure 30.4

The long stride and contralateral arm swing helps tension the whole body in angles appropriate for dealing with forces following heel-strike (front limb, note the forward swing of the opposite arm to help tension the posterior oblique sling), and for the preparation for toe-off on the posterior lower limb.

to tension elastic and contractile tissues optimally on the front and back legs during gait to minimize muscle work and its resultant metabolic cost. If overuse is a cause of soft tissue dysfunction, understanding how it should be optimally loaded is essential knowledge for any therapist as they strive to understand the complexities of the presenting client.

Developing a wider view of tissue and joint interactions is essential when considering soft

tissue dysfunctions; as described above, the foot, hip and spine are all interdependent for their appropriate loading. Joint interrelationships are particularly important during the retraining of healthy movement patterns, especially when using any forms of environmental intervention such as orthoses or changes in footwear, in which cases the therapist must consider the knock-on effects further along the chain.

Walking: The benefit of being on two legs

References

- Biewener, A. (2016) Locomotion as an emergent property of muscle contractile dynamics. *J Exp Biol.* 219: 285–294.
- Chaitow, L. & DeLany, J., 2008 *Clinical Application Of Neuromuscular Techniques*. Edinburgh: Churchill Livingstone Elsevier.
- Crompton, R., Vereecke, E. & Thorpe, S. (2008) Locomotion and posture from the common hominoid ancestor to fully modern hominins, with special reference to the last common panin/hominin ancestor. *Journal of Anatomy.* 212(4): 501–543.
- Fernández, P., Mongle, C., Leakey, L., Proctor, D., Orr, C., Patel, B., Almécija, S., Tocheri, M. and Jungers, W. (2018) Evolution and function of the hominin forefoot. *Proceedings of the National Academy of Sciences.* 115(35): 8746–8751.
- Fukunaga, T., Kawakami, Y., Kubo, K. & Kanehisa, H. (2002) Muscle and tendon interaction during human movements. *Exerc Sport Sci Rev.* 30: 106–110.
- Holowka, N. & Lieberman, D. (2018) Rethinking the evolution of the human foot: insights from experimental research. *J Exp Biol.* 221: 1–10
- Lieberman, D. (2012) Those feet in ancient times. *Nature.* 483: 550–551.
- McKeon, P., Hertel, J., Bramble, D. & Davis, I. (2014) The foot core system: a new paradigm for understanding intrinsic foot muscle function. *Br J Sports Med.* 49: 290.
- McNeill Alexander, R. (2002) Tendon elasticity and muscle function. *Comparative Biochemistry and Physiology Part A: Molecular & Integrative Physiology.* 133: 1001–1011.
- Osborn, H. (1928) The influence of bodily locomotion in separating man from the monkeys and apes. *Scientific Monthly*, 385–399.
- Perry, J. & Burnfield, J. (2010) *Gait Analysis*. Thorofare, NJ: SLACK.
- Pontzer, H. (2017) Economy and endurance in human evolution. *Curr Biol.* 27: R613–R621.
- Roberts, T. (2016) Contribution of elastic tissues to the mechanics and energetics of muscle function during movement. *J Exp Biol.* 219: 266–275.
- Rolf, I.P. (1989) *Rolfing: Reestablishing the natural alignment and structural integration of the human body for vitality and well-being*. Rochester, VT: Healing Arts Press.
- Rubenson, J., Pires, N., Loi, H., Piniger, G. & Shannon, D. (2012) On the ascent: the soleus operating length is conserved to the ascending limb of the force-length curve across gait mechanics in humans. *J Exp Biol.* 215: 3539–3551.
- Sawicki, G., Lewis, C. & Ferris, D. (2009) It pays to have a spring in your step. *Exerc Sport Sci Rev.* 37: 130–138.
- Silder, A., Whittington, B., Heiderscheit, B. & Thelen, D. (2007) Identification of passive elastic joint moment-angle relationships in the lower extremity. *J Biomech.* 40: 2628–2635.
- Vleeming, A., Schuenke, M., Masi, A., Carreiro, J., Danneels, L. & Willard, F. (2012) The sacroiliac joint: an overview of its anatomy, function and potential clinical implications. *Journal of Anatomy.* 221(6): 537–567.
- Vogel, S. & DeFerrari, A. (2013) *Comparative biomechanics*. Princeton, NJ: Princeton University Press.
- Whittington, B., Silder, A., Heiderscheit, B. & Thelen, D. (2008) The contribution of passive-elastic mechanisms to lower extremity joint kinetics during human walking. *Gait Posture.* 27: 628–634.
- Zorn, A. (2015) Elastic Walking. In: Schleip, R. & Baker, A. (eds), *Fascia in Sport and Movement*. Edinburgh, UK: Handspring.

